

3.4 Logarithmic Functions

Logarithmic functions arise naturally when we wish to determine an unknown power. They are widely used in engineering, science, economics and data analysis, particularly when modelling quantities that vary over many orders of magnitude.

Definition

The **logarithmic function** is written in the form

$$y = \log_a(x)$$

where:

- $a > 0$ is the base,
- $a \neq 1$,
- $x > 0$.

The logarithm answers the question:

“To what power must the base a be raised to obtain x ?”

In other words,

$$y = \log_a(x)$$

is equivalent to

$$a^y = x$$

3.4.1 Understanding Logarithms

Logarithmic and exponential statements are simply two different ways of expressing the same relationship.

Worked Example

Consider the exponential statement

$$2^4 = 16$$

This can be written in logarithmic form by taking log to the base 2 on both sides:

$$\log_2(2^4) = \log_2(16)$$

$$4 = \log_2(16)$$

This tells us that 2 must be raised to the power 4 to obtain 16.

Worked Example

Find the value of x :

$$3^x = 81$$

Writing the equation in logarithmic form, by taking log to the base 3 on both sides:

$$\log_3(3^x) = \log_3(81)$$

$$x = \log_3(81)$$

$$x = 4$$

3.4.2 Common and Natural Logarithms

Although logarithms can have many different bases, two are used most frequently.

Definition

The two most common logarithms are:

- The **common logarithm**

$$\log_{10}(x),$$

usually written simply as

$$\log(x)$$

- The **natural logarithm**

$$\log_e(x),$$

written as

$$\ln(x),$$

where

$$e \approx 2.71828$$

In engineering and science, the natural logarithm is used most frequently.

Worked Example

Evaluate the following using a calculator:

$$\log(1000)$$

Since

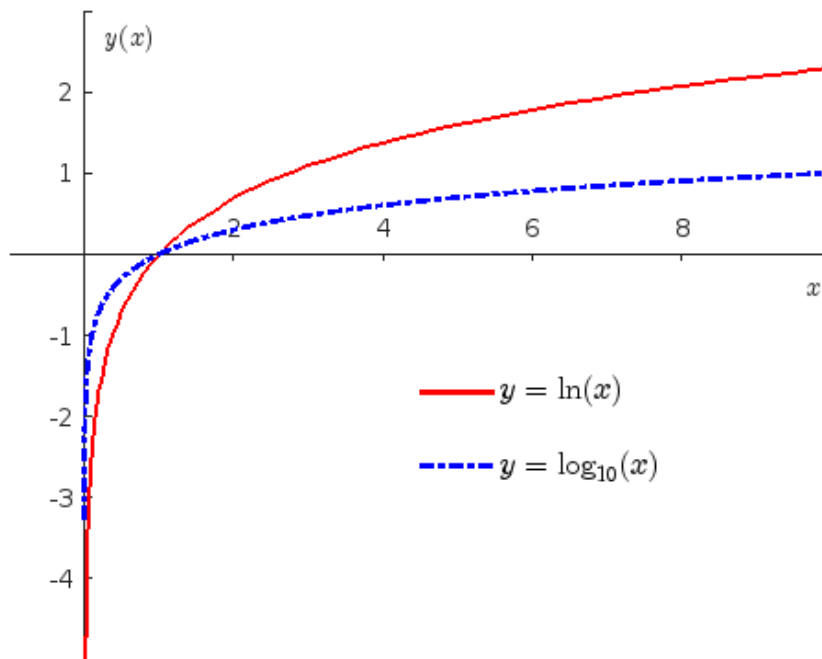
$$10^3 = 1000$$

we have

$$\log(1000) = 3$$

3.4.3 The Graph of a Logarithmic Function

The graph of a logarithmic function has a characteristic shape.



Definition

For the logarithmic function

$$y = \ln(x) \quad \text{and} \quad y = \log_{10}(x)$$

the following properties hold:

- The domain is $x > 0$.
- The graph passes through $(1, 0)$.
- There is no y -intercept.
- The y -axis is a vertical asymptote.
- The function increases as x increases.

More generally, logarithmic functions often appear in the form

$$y = A \ln(x) + B,$$

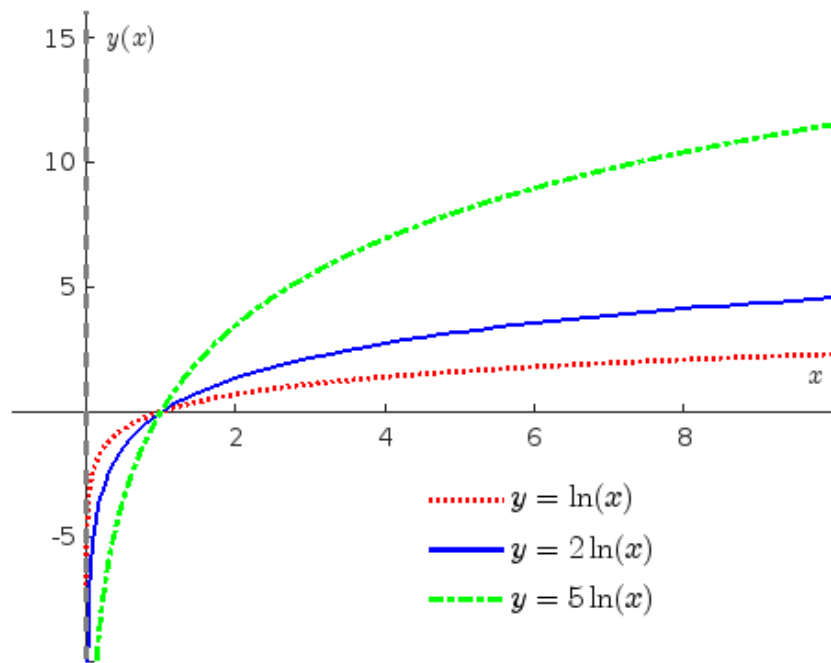
where A and B are constants.

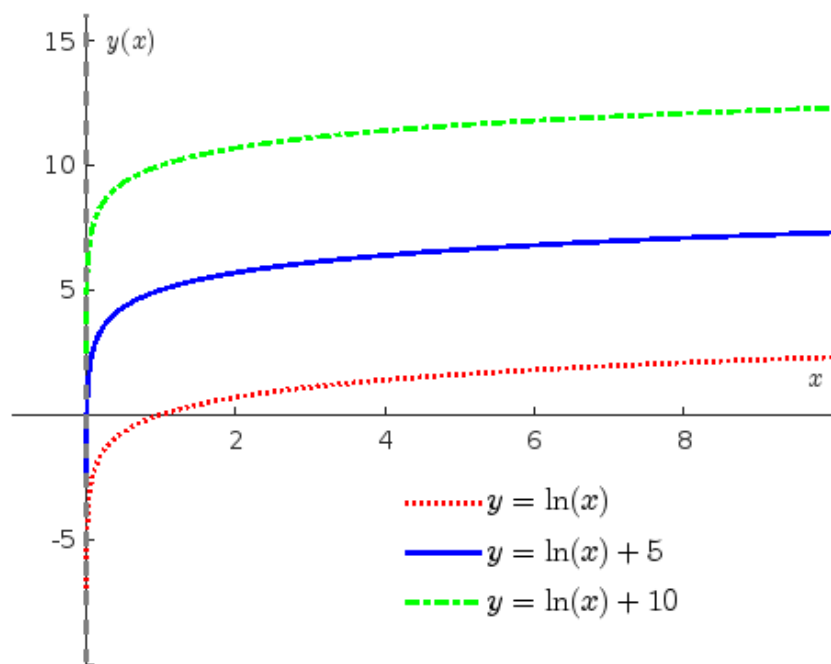
Definition

For

$$y = A \ln(x) + B$$

- Increasing $|A|$ stretches the graph vertically.
- If $A < 0$, the graph is reflected in the x -axis.
- The constant B shifts the graph vertically.





3.4.4 Laws of Logarithms

The laws of logarithms allow complicated expressions to be simplified.

Definition

For positive quantities A and B :

$$\log(A^n) = n \log(A)$$

$$\log(AB) = \log(A) + \log(B)$$

$$\log\left(\frac{A}{B}\right) = \log(A) - \log(B)$$

$$\log(1) = 0$$

Worked Example

Write the following as a single logarithm:

$$\log(6) + \log(3)$$

Using the product rule:

$$\begin{aligned}\log(6) + \log(3) &= \log(6 \times 3) \\ &= \log(18)\end{aligned}$$

Worked Example

Write the following as a single logarithm:

$$\ln(6) - \ln(3)$$

Using the quotient rule:

$$\begin{aligned}\ln(6) - \ln(3) &= \ln\left(\frac{6}{3}\right) \\ &= \ln(2)\end{aligned}$$

Worked Example

Write the following as a single logarithm:

$$2\log(x)$$

Using the power rule:

$$2\log(x) = \log(x^2)$$

3.4.5 Logarithms and Exponentials

Logarithmic functions and exponential functions are inverse functions.

Definition

The natural logarithmic and exponential functions undo each other.

$$\ln(e^x) = x$$

and

$$e^{\ln(x)} = x$$

Worked Example

Simplify $\ln(e^7)$.

Since logarithms and exponentials are inverse functions,

$$\ln(e^7) = 7$$

Worked Example

Simplify $e^{\ln(15)}$.

Since logarithms and exponentials are inverse functions,

$$e^{\ln(15)} = 15$$

3.4.6 Applications of Logarithms

Engineering Application

Logarithms are frequently used to solve equations where the unknown quantity appears as a power. For example,

$$2^x = 15$$

Taking logarithms gives

$$x = \log_2(15)$$

Without logarithms, equations of this form are often difficult to solve.

Engineering Application

Logarithmic scales are commonly used to measure quantities that vary over a very large range.

Logarithmic scales are used when values span a very large range because they compress large numbers while still showing smaller values clearly. This allows equal ratios (for example, 10 to 100 and 100 to 1000) to occupy equal distances on the scale, making patterns and trends easier to see. Examples include:

- Sound intensity (decibels),
- Earthquake magnitude,
- Signal processing,
- Information theory.

3.4.7 Exercises

Exercise

1. Rewrite the following exponential statements in logarithmic form:

(a) $2^5 = 32$

(b) $10^3 = 1000$

(c) $e^4 = 54.6$

2. Evaluate the following:

(a) $\log_2(64)$

(b) $\log_{10}(10000)$

(c) $\ln(e^6)$

3. Write each expression as a single logarithm:

(a) $\log(8) + \log(5)$

(b) $\ln(12) - \ln(4)$

(c) $3 \ln(x)$

4. Simplify the following:

(a) $\ln(e^{12})$

(b) $e^{\ln(20)}$

(c) $\ln(e^{2x})$

3.4 Solutions to Exercise 3.4.7

1. Rewrite the following exponential statements in logarithmic form

(a)

$$2^5 = 32$$

$$\log_2(2^5) = \log_2(32)$$

$$5 = \log_2(32)$$

(b)

$$10^3 = 1000$$

$$\log_1 0(10^3) = \log_1 0(1000)$$

$$3 = \log_1 0(1000)$$

(c)

$$e^4 = 54.6$$

$$\ln(e^4) = \ln(54.6)$$

$$4 = \ln(54.6)$$

2. Evaluate the following:

(a)

$$\log_2(64) = 6$$

(b)

$$\log_{10}(10000) = 4$$

(c)

$$\ln(e^6) = 6$$

3. Write each expression as a single logarithm:

(a) $\log(8) + \log(5)$. Using the law $\log(A) + \log(B) = \log(AB)$, we obtain

$$\log(8) + \log(5) = \log(8 \times 5)$$

$$= \log(40)$$

(b) $\ln(12) - \ln(4)$. Using the law $\log(A) - \log(B) = \log\left(\frac{A}{B}\right)$, we obtain

$$\ln(12) - \ln(4) = \ln\left(\frac{12}{4}\right)$$

$$= \ln(3)$$

(c) $3 \ln(x)$. Using the law $n \log(A) = \log(A^n)$, we obtain

$$3 \ln(x) = \ln(x^3)$$

4. Simplify the following:

(a) $\ln(e^{12})$. Since logarithms and exponentials are inverse functions,

$$\ln(e^{12}) = 12$$

(b) $e^{\ln(20)}$. Since logarithms and exponentials are inverse functions,

$$e^{\ln(20)} = 20$$

(c) $\ln(e^{2x})$. Since logarithms and exponentials are inverse functions,

$$\ln(e^{2x}) = 2x$$